

An Introduction to Tabular Foundation Models

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Tabular foundation models

- *TabPFN* (Hollmann et al. 2023)
- *TabICL* (QU et al. 2025)

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In-context learning setup (tabular data $\{(X_i, Y_i)\}_{1 \leq i \leq n}$)

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- ① split range of Y into 1000 classes

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- ② pass entire dataset to a transformer

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- 1 split range of Y into 1000 classes
- 2 pass entire dataset to a transformer
- 3 classify rows and compute avg. loss

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Repeat for millions of (usually synthetic) datasets.

Tabular foundation models

Architecture (transformer-based)

X_1	X_2	Y
$x_1^{(1)}$	$x_2^{(1)}$	$y^{(1)}$
$x_1^{(2)}$	$x_2^{(2)}$	$y^{(2)}$
$x_1^{(3)}$	$x_2^{(3)}$	$y^{(3)}$
$x_1^{(4)}$	$x_2^{(4)}$	

tabular dataset

Tabular foundation models

Architecture (transformer-based)

- ① embed each cell

$(4 \times 2 \times d)$

$(3 \times 1 \times d)$

X_1	X_2	Y
$\bar{x}_1^{(1)}$	$\bar{x}_2^{(1)}$	$\bar{y}^{(1)}$
$\bar{x}_1^{(2)}$	$\bar{x}_2^{(2)}$	$\bar{y}^{(2)}$
$\bar{x}_1^{(3)}$	$\bar{x}_2^{(3)}$	$\bar{y}^{(3)}$
$\bar{x}_1^{(4)}$	$\bar{x}_2^{(4)}$	

tabular dataset

Tabular foundation models

Architecture (transformer-based)

- 1 embed each cell
 $(4 \times 2 \times d)$
 $(3 \times 1 \times d)$
- 2 column-attention

X_1	X_2	Y
$\bar{x}_1^{(1)}$	$\bar{x}_2^{(1)}$	$\bar{y}^{(1)}$
$\bar{x}_1^{(2)}$	$\bar{x}_2^{(2)}$	$\bar{y}^{(2)}$
$\bar{x}_1^{(3)}$	$\bar{x}_2^{(3)}$	$\bar{y}^{(3)}$
$\bar{x}_1^{(4)}$	$\bar{x}_2^{(4)}$	

tabular dataset

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Architecture (transformer-based)

- 1 embed each cell
($4 \times 2 \times d$)
($3 \times 1 \times d$)
- 2 column-attention
- 3 add labels to features

X_1	X_2	Y
$\bar{x}_1^{(1)}$	$\bar{x}_2^{(1)}$	$\bar{y}^{(1)}$
$\bar{x}_1^{(2)}$	$\bar{x}_2^{(2)}$	$\bar{y}^{(2)}$
$\bar{x}_1^{(3)}$	$\bar{x}_2^{(3)}$	$\bar{y}^{(3)}$
$\bar{x}_1^{(4)}$	$\bar{x}_2^{(4)}$	

tabular dataset

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Architecture (transformer-based)

- 1 embed each cell
 $(4 \times 2 \times d)$
 $(3 \times 1 \times d)$
- 2 column-attention
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- 4 row-attention

X_1	X_2	Y
$\bar{x}_1^{(1)}$	$\bar{x}_2^{(1)}$	$\bar{y}^{(1)}$
$\bar{x}_1^{(2)}$	$\bar{x}_2^{(2)}$	$\bar{y}^{(2)}$
$\bar{x}_1^{(3)}$	$\bar{x}_2^{(3)}$	$\bar{y}^{(3)}$
$\bar{x}_1^{(4)}$	$\bar{x}_2^{(4)}$	

tabular dataset

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Architecture (transformer-based)

- 1 embed each cell
 $(4 \times 2 \times d)$
 $(3 \times 1 \times d)$
- 2 column-attention
- 3 add labels to features
- 4 row-attention
- 5 predict (classify) Y

(unlabeled cells are not attended to, and can only attend to labeled ones and themselves)

X_1	X_2	Y
$\bar{x}_1^{(1)}$	$\bar{x}_2^{(1)}$	$\bar{y}^{(1)}$
$\bar{x}_1^{(2)}$	$\bar{x}_2^{(2)}$	$\bar{y}^{(2)}$
$\bar{x}_1^{(3)}$	$\bar{x}_2^{(3)}$	$\bar{y}^{(3)}$
$\bar{x}_1^{(4)}$	$\bar{x}_2^{(4)}$	$\hat{y}^{(4)}$

tabular dataset

Thank you!

References I

- [1] Noah Hollmann, Samuel Müller, Katharina Eggersperger, and Frank Hutter. “TabPFN: A Transformer That Solves Small Tabular Classification Problems in a Second”. In: *The Eleventh International Conference on Learning Representations*. 2023 (cit. on pp. 2–8).
- [2] Jingang QU, David Holzmüller, Gaël Varoquaux, and Marine Le Morvan. “TabICL: A Tabular Foundation Model for In-Context Learning on Large Data”. In: *Forty-second International Conference on Machine Learning*. 2025 (cit. on pp. 2–8).